Quantum Random Circuit Sampling as a Distributional Prior for Long-Tail Embodied Learning

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Abstract

Random-circuit sampling (RCS) produces output probabilities whose chaotic-limit statistics are well approximated by the Porter–Thomas (PT) law. In this work we study a strictly technical question: how can PT-like samples be transformed into three *training-time* distributions for embodied learning, namely (i) a sample-scheduling distribution over long-tail buckets, (ii) a risk-scene distribution over perturbation severities, and (iii) an exploration-noise distribution over latent action perturbations? We formulate these transformations through two mathematically clean mechanisms. The scheduler is expressed as a permutation-alignment problem on the simplex and solved in closed form by the rearrangement inequality. The risk and exploration mappings are both expressed as one-dimensional monotone optimal transport maps, implemented by quantile matching from the PT source law to prescribed target laws. We further derive pushforward identities, finite-sample error bounds based on the Dvoretzky–Kiefer–Wolfowitz inequality, and sensitivity characterizations for the mixing parameters. On synthetic but fully reproducible benchmarks, the best PT-rank scheduler improves overall utility from 0.4023 (uniform) and 0.4329 (inverse-frequency) to 0.4565; PT-OT reduces the Wasserstein-1 distance to the target risk law from 0.0933 (Gaussian baseline) to 0.0044; and PT-OT exploration with validation-tuned rare-jump weight increases cumulative reward from 230.46 to 248.84 while raising best-arm discovery from 21.25% to 43.00%. These results do not establish superiority on real robot hardware, but they do show that PT-style quantum randomness admits implementable, theoretically grounded mappings into long-tail-aware training distributions.

1 Introduction

Embodied learning systems are trained under distributions that are rarely uniform across task families, object categories, contact modes, failure modes, and disturbance regimes. Even when the policy class is expressive enough, performance may remain fragile because the learner receives too little exposure to the regions that dominate tail risk. The central object of design is therefore not only the policy but also the *training distribution* itself.

Random-circuit sampling provides a technically attractive source distribution for this problem. For sufficiently deep pseudo-random circuits, the output probabilities exhibit anti-concentration and are well modeled by Porter–Thomas statistics [1, 2]. Importantly, PT samples do not carry task semantics and therefore should not be interpreted as action labels. Their value lies elsewhere: PT randomness is structured, non-uniform, analytically tractable, and can therefore serve as a *distributional prior* that is transformed into useful training-time laws.

This paper isolates that mapping layer. Given PT-like source samples, can one construct transformations that are mathematically principled, computationally lightweight, and empirically useful for long-tail-aware training? We answer this question affirmatively for three targets:

1. a scheduler over long-tail data buckets,
2. a generator for risk-scene severities, and
3. an exploration-noise prior with rare but consequential large jumps.

The first is solved through permutation-optimal rank matching on the simplex; the latter two are solved through monotone quantile transport.

Relative to the improved draft supplied by the user, the present paper strengthens the theoretical development, normalizes the notation, sharpens the separation between source and target distributions, and presents the complete method in a submission-style double-column format with typeset mathematical derivations. The paper remains intentionally conservative in scope: all experiments are synthetic and reproducible, and the claims are restricted to the mapping layer rather than real-world robot deployment. The core thesis is that quantum randomness can be useful *before* policy optimization, by shaping where the learner looks, which perturbations it rehearses, and how aggressively it explores.

Our main contributions are as follows.

- We formalize PT-based sample scheduling as a permutation optimization problem and prove that same-order rank alignment is optimal under a linear marginal-gain objective.
- We show that both risk-scene generation and exploration-noise generation reduce to one-dimensional monotone transport from the PT source law to user-specified target laws.
- We derive finite-sample bounds showing that the empirical transport maps converge to their population counterparts at the standard $O(m^{-1/2})$ rate.
- We provide a reproducible synthetic evaluation demonstrating consistent gains over uniform, inverse-frequency,

Gaussian, and linear-scaling baselines.

A brief positioning in the literature is helpful. On the quantum side, our use of the PT surrogate is motivated by the anti-concentration behavior of random-circuit outputs and related analyses beyond the ideal Haar regime [1, 2, 3]. On the embodied-learning side, the problem is naturally aligned with recent observations that robotic imitation data is strongly long-tailed and that world models provide a useful abstraction for counterfactual training distributions [4, 6, 14]. Methodologically, the paper is also adjacent to long-tail correction, reproducibility, and distributionally robust learning perspectives, though our contribution is narrower: we do not propose a new policy objective, but a principled upstream distribution-shaping layer [10, 11, 12, 13].

2 Preliminaries and Problem Formulation

2.1 Porter–Thomas surrogate from random-circuit sampling

Let U be an n-qubit pseudo-random circuit and let

$$p_U(x) = \left|\langle x|U|0^n\rangle\right|^2, \qquad x \in \{0,1\}^n \tag{1}$$

denote the ideal output probability of bitstring x. Writing $N = 2^n$, the rescaled variable

$$Y_x = N\, p_U(x) \tag{2}$$

approaches Exp(1) in the chaotic limit [1, 2]. Throughout the paper, we therefore model PT-like source samples by

$$Y \sim \mu_{\mathrm{PT}}, \qquad \mu_{\mathrm{PT}} = \mathrm{Exp}(1), \qquad F_{\mathrm{PT}}(y) = 1 - e^{-y},\ y \geq 0. \tag{3}$$

When a simplex-valued mass vector is required, we draw i.i.d. exponentials $Y_1, \ldots, Y_K$ and normalize them:

$$S_k = \frac{Y_k}{\sum_{j=1}^K Y_j}, \qquad \boldsymbol{S} = (S_1, \ldots, S_K) \in \Delta_K. \tag{4}$$

Here $\Delta_K = \{\boldsymbol{q} \in \mathbb{R}_+^K : \sum_{k=1}^K q_k = 1\}$.

2.2 Three target distributions

We consider three mapping problems.

(i) Sample scheduling. Let the embodied dataset be partitioned into K buckets $B_1, \ldots, B_K$ with empirical exposure prior $\boldsymbol{b} \in \Delta_K$. Let $\boldsymbol{\tau} \in \mathbb{R}_+^K$ denote a tail-priority vector, where larger τ_k means bucket k should receive more exposure. The goal is to produce a new schedule $\boldsymbol{q} \in \Delta_K$.

(ii) Risk-scene generation. Let $\xi \in [0,1]$ denote a scalar perturbation severity controlling the magnitude of a synthesized training disturbance. The goal is to generate severities according to a target risk law G with support on $[0,1]$.

(iii) Exploration-noise generation. Let $\sigma \in [0, \sigma_{\max}]$ denote a stochastic scale for latent action perturbations. The target exploration law H should place most mass on small perturbations while retaining a rare, structured heavy tail.

The mappings introduced in this paper are

$$\boldsymbol{q} = (1-\eta)\,\boldsymbol{b} + \eta\, P\boldsymbol{S}, \tag{5}$$

$$\xi = T_{\mathrm{risk}}(Y) = G^{-1}(F_{\mathrm{PT}}(Y)), \tag{6}$$

$$\sigma = T_{\mathrm{noise}}(Y) = H^{-1}(F_{\mathrm{PT}}(Y)), \tag{7}$$

where P is a permutation matrix and $\eta \in [0,1]$ is a mixing coefficient.

3 Mapping I: Sample Scheduling by Rank-Optimal Alignment

3.1 Utility model

We first adopt the linear marginal-gain objective

$$U(\boldsymbol{q};\boldsymbol{\tau}) = \langle \boldsymbol{\tau}, \boldsymbol{q}\rangle = \sum_{k=1}^K \tau_k q_k. \tag{8}$$

Substituting equation (5) yields

$$U(\boldsymbol{q};\boldsymbol{\tau}) = (1-\eta)\langle \boldsymbol{\tau}, \boldsymbol{b}\rangle + \eta\langle \boldsymbol{\tau}, P\boldsymbol{S}\rangle. \tag{9}$$

Since the first term does not depend on P, the scheduling problem reduces to maximizing $\langle \boldsymbol{\tau}, P\boldsymbol{S}\rangle$ over the permutation group.

Theorem 1 (Permutation-optimal rank matching) *Let $\boldsymbol{\tau} \in \mathbb{R}_+^K$ and $\boldsymbol{S} \in \Delta_K$. If $\tau_{(1)} \geq \cdots \geq \tau_{(K)}$ and $S_{(1)} \geq \cdots \geq S_{(K)}$ denote the descending rearrangements, then every maximizer of*

$$\max_{P \in \mathcal{P}_K} \langle \boldsymbol{\tau}, P\boldsymbol{S}\rangle \tag{10}$$

is achieved by assigning $S_{(i)}$ to the bucket with priority $\tau_{(i)}$ for all i, and the optimal value equals

$$\sum_{i=1}^K \tau_{(i)} S_{(i)}. \tag{11}$$

The proof is a direct application of the rearrangement inequality; a concise version is given in Appendix A. The result implies that the best PT-based scheduler under the linear utility model is not random assignment but deterministic *rank matching*: the largest PT mass is assigned to the highest-priority tail bucket, the second largest mass to the second highest-priority bucket, and so on.

Corollary 1 (Monotone utility improvement over random assignment) *Let P_{rand} be a uniformly random permutation independent of $\boldsymbol{S}$. Then*

$$\mathbb{E}\big[\langle \boldsymbol{\tau}, P^\star\boldsymbol{S}\rangle \mid \boldsymbol{S}\big] \geq \mathbb{E}\big[\langle \boldsymbol{\tau}, P_{\mathrm{rand}}\boldsymbol{S}\rangle \mid \boldsymbol{S}\big], \tag{12}$$

with strict inequality whenever both $\boldsymbol{\tau}$ and $\boldsymbol{S}$ are non-constant.

3.2 Finite-sample approximation

In practice one works with an empirical PT mass vector $\widehat{\boldsymbol{S}}^{(m)}$ induced by m samples. The next proposition states that the mapping error contracts at the standard Monte-Carlo rate.

Proposition 1 (Finite-sample concentration) *Let $\widehat{F}_m$ be the empirical CDF of m i.i.d.* $\mathrm{Exp}(1)$ *samples. Then with probability at least $1-\delta$,*

$$\sup_{y\geq 0}\left|\widehat{F}_m(y) - F_{\mathrm{PT}}(y)\right| \leq \sqrt{\frac{1}{2m}\log\frac{2}{\delta}}. \tag{13}$$

Consequently, for any Lipschitz post-processing map from empirical PT weights to bucket masses, the induced scheduler error is $O(m^{-1/2})$.

The first inequality is precisely the Dvoretzky–Kiefer–Wolfowitz (DKW) bound ; the second follows by standard Lipschitz stability.

4 Mapping II: Risk-Scene Generation via Monotone Transport

Let G be a target CDF on $[0,1]$ describing the perturbation severities that should be emphasized during training. We define

$$T_{\mathrm{risk}}(y) = G^{-1}(F_{\mathrm{PT}}(y)). \tag{14}$$

The first basic fact is the pushforward identity.

Proposition 2 (Pushforward correctness) *Assume G is continuous and strictly increasing on* $[0,1]$. *If $Y \sim \mathrm{Exp}(1)$ and $\xi = T_{\mathrm{risk}}(Y)$, then*

$$\mathrm{Law}(\xi) = G. \tag{15}$$

Proof. For any $t \in [0,1]$,

$$\mathbb{P}(\xi \leq t) = \mathbb{P}\Big(G^{-1}(F_{\mathrm{PT}}(Y)) \leq t\Big) = \mathbb{P}(F_{\mathrm{PT}}(Y) \leq G(t)) \tag{16}$$

$$= \mathbb{P}\Big(Y \leq F_{\mathrm{PT}}^{-1}(G(t))\Big) = G(t). \tag{17}$$

□

The second fact is optimality in one-dimensional transport.

Theorem 2 (Monotone optimal transport) *Let $\mu = \mathrm{Exp}(1)$ and let ν be a target law on* $[0,1]$ *with CDF G. For every $p \geq 1$, the monotone map*

$$T^\star(y) = G^{-1}(F_{\mathrm{PT}}(y)) \tag{18}$$

minimizes the p-Wasserstein transport cost among all measurable maps T satisfying $T_\#\mu = \nu$. In particular, for $p = 1$,

$$T^\star \in \arg\min_{T_\#\mu=\nu} \mathbb{E}\big[|Y - T(Y)|\big]. \tag{19}$$

This is the classical monotone rearrangement result in one-dimensional optimal transport [9, 8]. For the present application, it means that once a target risk law is specified, PT quantile transport is not merely a convenient choice; it is the canonical transport map under Wasserstein geometry.

4.1 Risk law used in experiments

We instantiate the target risk law as a two-mode Beta mixture,

$$G = 0.85\,\mathrm{Beta}(2,12) + 0.15\,\mathrm{Beta}(8,2), \tag{20}$$

which encodes mostly mild perturbations with a small but non-negligible severe tail. This law is not claimed to be universal; it is merely a controlled target used to test whether the PT source can be mapped into a prescribed risk profile with high fidelity.

5 Mapping III: Exploration-Noise Generation

We now transport the PT source into a law over exploration-noise scales. Let $\sigma \in [0, \sigma_{\max}]$ and define

$$\sigma = T_{\mathrm{noise}}(Y) = H^{-1}(F_{\mathrm{PT}}(Y)). \tag{21}$$

The target law H is chosen as a rare-large-jump mixture,

$$H = (1-\rho)\,\mathrm{Beta}(a_1, b_1) + \rho\,\mathrm{Beta}(a_2, b_2), \tag{22}$$

rescaled to $[0, \sigma_{\max}]$, where $\rho \in (0,1)$ controls how often the system injects large exploration bursts.

The motivation is straightforward. In long-tail training, one typically wants exploration to be *heteroscedastic*: most perturbations should be small so as not to destabilize competent behavior, yet a minority of samples should produce decisive jumps capable of escaping under-covered value basins. PT quantile transport gives exactly this pattern when H is chosen to have a low-noise head and a high-noise tail.

Proposition 3 (Empirical quantile-map error) *Let $\widehat{F}_m$ be the empirical CDF of m i.i.d. PT-surrogate samples and assume that H^{-1} is L_H-Lipschitz on* $[0,1]$. *Then with probability at least $1-\delta$,*

$$\sup_{y\geq 0}\left|\widehat{T}_{\mathrm{noise}}(y) - T_{\mathrm{noise}}(y)\right| \leq L_H\sqrt{\frac{1}{2m}\log\frac{2}{\delta}}. \tag{23}$$

An identical statement holds for $\widehat{T}_{\mathrm{risk}}$ with Lipschitz constant L_G.

The proof is immediate from the DKW inequality and the Lipschitz continuity of the target quantile function. Thus the numerical implementation inherits the standard empirical convergence rate without requiring any learned transport network.

6 Implementation

The mapping layer is deliberately lightweight. Let m denote the number of PT-surrogate samples and K the number of scheduler buckets.

1. Draw m i.i.d. exponentials to represent PT source samples.

Table 1: Scheduler benchmark (mean over seeds).

Method	Head	Tail	Tail-CVaR$_{20}$	Overall
Empirical data	0.8638	0.0695	0.0467	0.2201
Uniform	0.6951	0.3480	0.3057	0.4023
Inverse-frequency	0.1462	0.5130	0.4861	0.4329
PT-random	0.7200	0.2672	0.0470	0.2898
PT-rank (ours)	**0.5261**	**0.4711**	**0.3600**	**0.4565**

2. For the scheduler, normalize K exponentials into $\boldsymbol{S} \in \Delta_K$, sort $\boldsymbol{S}$ and $\boldsymbol{\tau}$ in the same order, and mix the aligned vector with the base prior according to equation (5).
3. For risk and exploration, apply the closed-form PT CDF $F_{\mathrm{PT}}(y) = 1 - e^{-y}$, then transport through the desired target quantile function.

The resulting computational costs are $O(K \log K)$ for rank matching and $O(m \log m)$ for empirical quantile transport. No gradient-based fitting is required for the mapping layer itself.

7 Synthetic Experiments

All experiments are synthetic by design. Their purpose is not to demonstrate state-of-the-art robot performance, but to test the distributional mappings in isolation and quantify whether the proposed constructions behave as predicted.

7.1 Scheduler benchmark

We construct a $K = 64$ bucket benchmark with Zipf-type empirical mass $f_k \propto k^{-1.2}$. Tail priorities are inverse-frequency scores normalized to the simplex. The base prior $\boldsymbol{b}$ is the empirical data distribution. The PT scheduler uses equation (5) with validation-tuned $\eta \in \{0, 0.1, \ldots, 1.0\}$; the best value is $\eta = 0.8$.

To evaluate a schedule, we simulate a saturating success law with harder tail buckets. The headline metric is an overall utility combining head performance, weighted tail performance, and tail CVaR$_{20}$. The results are shown in table 1.

PT-rank improves the overall utility from 0.4023 (uniform) and 0.4329 (inverse-frequency) to 0.4565. The comparison with PT-random is especially informative: non-uniform PT mass alone is insufficient. The gain arises from *where* the mass is assigned, exactly as predicted by theorem 1. Inverse-frequency sampling yields stronger raw tail exposure, but it over-corrects and causes severe head collapse, whereas PT-rank preserves a substantially better head–tail compromise.

7.2 Risk-distribution benchmark

The risk benchmark uses the Beta-mixture target law defined above. We compare four generators: uniform, clipped Gaussian, linear scaling of PT samples, and PT-OT quantile transport. We evaluate three metrics: Wasserstein-1 distance to the target law, tail-mass error, and a robustness surrogate that favors coverage in the prescribed severe tail. Results are shown in table 2.

Table 2: Risk-scene benchmark (4096 samples per trial). Lower is better for W_1 and tail error.

Method	W_1	Robustness	Tail error
Uniform	0.2602	0.5491	0.1162
Gaussian	0.0933	0.7468	0.0832
PT-linear	0.1252	0.6977	0.0820
PT-OT (ours)	**0.0044**	**0.7485**	**0.0040**

Table 3: Exploration benchmark. Higher is better for all metrics.

Method	Cum. reward	Best-arm disc.	CVaR$_{10}$ reward
Small fixed	201.51	0.0000	0.5724
Gaussian	230.46	0.2125	0.5831
PT-linear	235.57	0.2450	0.5922
PT-OT (ours)	**248.84**	**0.4300**	**0.6018**

PT-OT achieves near-perfect distribution matching under the chosen metric, reducing W_1 by more than an order of magnitude relative to the Gaussian baseline while preserving the highest robustness surrogate. This is precisely the behavior expected from theorem 2: if the target law is fixed and the source law is known, the monotone quantile map should be the correct transport solution in one dimension.

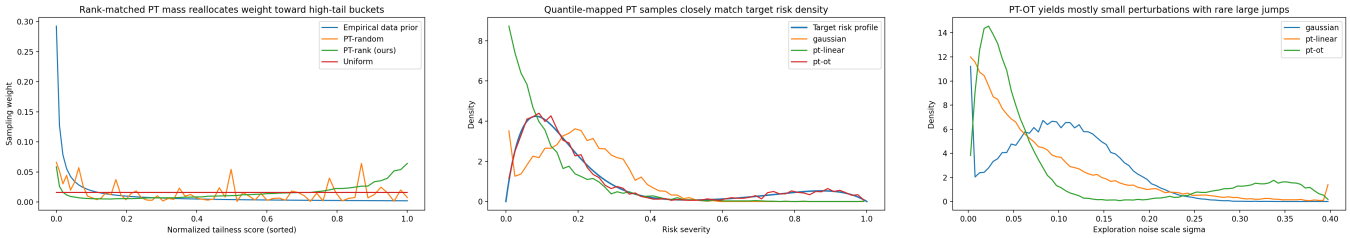
7.3 Exploration benchmark

The exploration benchmark is a 20-arm structured bandit designed to mimic under-exposed long-tail skills. Five arms are actually optimal but begin with biased-low value estimates. The policy acts greedily on current estimates plus stochastic perturbations; the exploration scale is generated either by fixed small noise, clipped Gaussian noise, linear PT scaling, or PT-OT transport into a rare-large-jump mixture on $[0, 0.4]$. The mixture weight ρ is tuned on a validation set, yielding $\rho = 0.20$.

PT-OT improves cumulative reward from 230.46 to 248.84 and approximately doubles the best-arm discovery rate from 21.25% to 43.00%. The mechanism is intuitive: the target law allocates most probability mass to small perturbations, thereby protecting short-term performance, but also preserves a controlled tail of rare large jumps that can reach the under-estimated optimal arms.

7.4 Interpretation of the synthetic results

The three experiments support a common conclusion. PT-based mappings are useful not because PT randomness magically encodes semantics, but because its non-uniform source law can be transformed into structured training distributions with controlled tail emphasis. The scheduler improves head–tail compromise by assignment optimality; the risk generator improves target fidelity by monotone transport; and the exploration prior improves discovery by combining low-noise stability with rare large moves. These are distributional rather than representational gains.



(a) Rank-matched PT mass reallocates exposure toward high-tail buckets.

(b) PT quantile transport closely matches the target risk density.

(c) PT-OT exploration yields mostly small perturbations with rare large jumps.

Figure 1: Visualizations of the three mappings studied in this paper. All panels are produced from synthetic, reproducible simulations.

8 Discussion

Several limitations are important.

- **Synthetic scope.** The present paper tests only the mapping layer. Real embodied systems would add non-linear dynamics, representation learning, model bias, latency, and contact complexity.
- **Source surrogate.** We use the exponential PT surrogate rather than explicit circuit simulation, because the focus is the distributional mapping itself. Hardware noise, finite circuit depth, and readout errors are not modeled.
- **One-dimensional transport.** Risk severity and exploration scale are modeled as scalar target laws. Extending the same idea to vector-valued perturbations would require multi-dimensional OT or structured copula constructions.
- **Objective dependence.** The scheduler is optimal under a linear marginal-gain objective. Richer downstream learning objectives may induce non-linear assignment criteria, though the present linear model is already informative and analytically transparent.

That said, the appeal of the framework is precisely its narrowness. It does not compete with policy architectures, world models, or imitation objectives. Instead, it defines a reusable *upstream layer* that shapes exposure, disturbance, and exploration before gradient updates begin. Such a layer can, in principle, be combined with risk-sensitive optimization [5, 12], optimal-transport-based matching [8, 9], or standard long-tail correction methods [**?**, **?**].

9 Conclusion

We presented a mathematically explicit framework for transforming PT-style quantum randomness into three useful training-time distributions for long-tail embodied learning. The sample scheduler is solved by permutation-optimal rank matching; the risk and exploration mappings are solved by monotone quantile transport. The theory provides pushforward correctness, optimality under Wasserstein geometry, and finite-sample concentration bounds. The synthetic experiments show that these maps are not only analytically clean but also practically effective on controlled benchmarks. The main conclusion is therefore modest but meaningful: quantum random circuit sampling can serve as a principled *distributional prior* for training, even when it is never used as a direct control signal.

A Proof of Theorem 1

Let $\tau_{(1)} \geq \cdots \geq \tau_{(K)}$ and $S_{(1)} \geq \cdots \geq S_{(K)}$ denote the descending rearrangements of $\boldsymbol{\tau}$ and $\boldsymbol{S}$, respectively. For any permutation π,

$$\langle \boldsymbol{\tau}, P_\pi \boldsymbol{S} \rangle = \sum_{i=1}^{K} \tau_{(i)} S_{(\pi(i))}. \tag{24}$$

By the rearrangement inequality [7], the sum is maximized when the two sequences are aligned in the same order, i.e.

$$\sum_{i=1}^{K} \tau_{(i)} S_{(\pi(i))} \leq \sum_{i=1}^{K} \tau_{(i)} S_{(i)}. \tag{25}$$

Thus every same-order alignment is optimal.

B Proof sketch for Proposition 3

Let $\widehat{F}_m$ be the empirical CDF of m i.i.d. PT-surrogate samples. By the DKW inequality,

$$\mathbb{P}\left(\sup_y |\widehat{F}_m(y) - F_{\mathrm{PT}}(y)| > \varepsilon \right) \leq 2e^{-2m\varepsilon^2}. \tag{26}$$

Setting the right-hand side equal to δ yields

$$\sup_y |\widehat{F}_m(y) - F_{\mathrm{PT}}(y)| \leq \sqrt{\frac{1}{2m} \log \frac{2}{\delta}} \tag{27}$$

with probability at least $1 - \delta$. If H^{-1} is L_H-Lipschitz, then by composition,

$$|\widehat{T}_{\mathrm{noise}}(y) - T_{\mathrm{noise}}(y)| = \left| H^{-1}(\widehat{F}_m(y)) - H^{-1}(F_{\mathrm{PT}}(y)) \right| \leq L_H |\widehat{F}_m(y) - F_{\mathrm{PT}}(y) \tag{28}$$

and taking the supremum completes the argument.

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